

From qubits to logic

Engineering Fault-Tolerant Quantum Systems

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Executive Summary

Fault-tolerant quantum computing (FTQC) is the critical milestone where quantum machines move from fragile research prototypes into reliable, utility-scale computers. By suppressing errors through quantum error correction, FTQC allows for scalable systems capable of solving problems beyond the reach of classical supercomputers, enabling applications from materials and drug discovery to optimization and cryptography.

FTQC is moving from theory to engineering reality, with advancements across multiple platforms. Google's Willow processor demonstrated exponential error suppression below the surface-code threshold¹, Quantinuum's trapped-ion system achieved break-even quantum error correction with logical gates outperforming physical ones², and neutral-atom arrays scaled beyond 6,000 qubits with multi-second coherence³. Collectively, these results confirm that the field has entered a phase where logical qubits can consistently outperform physical ones, which is an essential milestone for scalable FTQC.

Different modalities bring complementary strengths and challenges. Superconducting circuits lead in fabrication maturity and gate speed but face wiring and cryogenic bottlenecks at scale. Trapped ions deliver record fidelities and long-range connectivity, though their scaling is limited by mode crowding and laser control.

Neutral atoms achieve unparalleled scale and reconfigurability but are still working to improve two-qubit fidelity and gate speed. Spin qubits offer density and CMOS compatibility but must overcome fabrication variability. Photonic qubits, operating at room temperature and ideal for networking, are constrained by photon loss and probabilistic entanglement.

Atom-photon architectures aim to eliminate probabilistic bottlenecks, enabling modular and distributed quantum systems. Over the next decade, logical qubits are expected to consistently outperform physical qubits, and million-qubit systems will become possible. For investors and policymakers, the implication is clear: strategic support for diverse hardware approaches can catalyze the transition from today's prototypes to practical, error-corrected quantum machines, with impact across science, industry, and national competitiveness. For industry leaders and academic researchers, the message is equally direct: sustained investment in scalable architectures, cross-disciplinary collaboration, and co-designed hardware-software error-correction strategies will determine who captures their commercial and scientific rewards.

In this white paper, we assess the state of each major modality, as demonstrated by key players, under a unified framework, covering operating principles, performance, QEC approaches, and scaling outlook, and examine how hybrid atom-photon architecture addresses key limitations of photonic FTQC.

“For more than two decades, the theoretical foundations of quantum error correction have been developed, converging on the requirements for building a fault-tolerant quantum computer. In recent years, the first functional logical elements have been experimentally demonstrated across a broad range of hardware platforms, showing steadily improving performance and marking real progress toward the fault-tolerant era. We’re not there yet—but the future is coming into focus.

Michael Slutsky

Head of Theory at Quantum Source

”

“A fault-tolerant quantum computer capable of performing economically valuable applications requires at least 100 high-quality logical qubits that can execute algorithms which are at least millions of operations deep. Such a platform requires even more numerous physical qubits as well as key enabling capabilities including mid-circuit measurement, logical branching, all-to-all connectivity, better-than-threshold fidelities, and long coherence times. We would advise decision-makers to look at companies which are actively on a path to providing all of these building blocks. Additionally, these companies should have already demonstrated performing algorithms with logical qubits and have a path to rapidly scale their quality and number of logical qubits.

Dr. Ben Bloom

Founder & CEO at Atom Computing

”

Introduction

As quantum computing advances toward practical applications, fault tolerance has emerged as the decisive benchmark for long-term viability. While today's noisy intermediate-scale quantum (NISQ) systems are beginning to demonstrate value in specialized domains, truly reliable quantum computation will require architectures that are capable of suppressing and managing errors at scale.

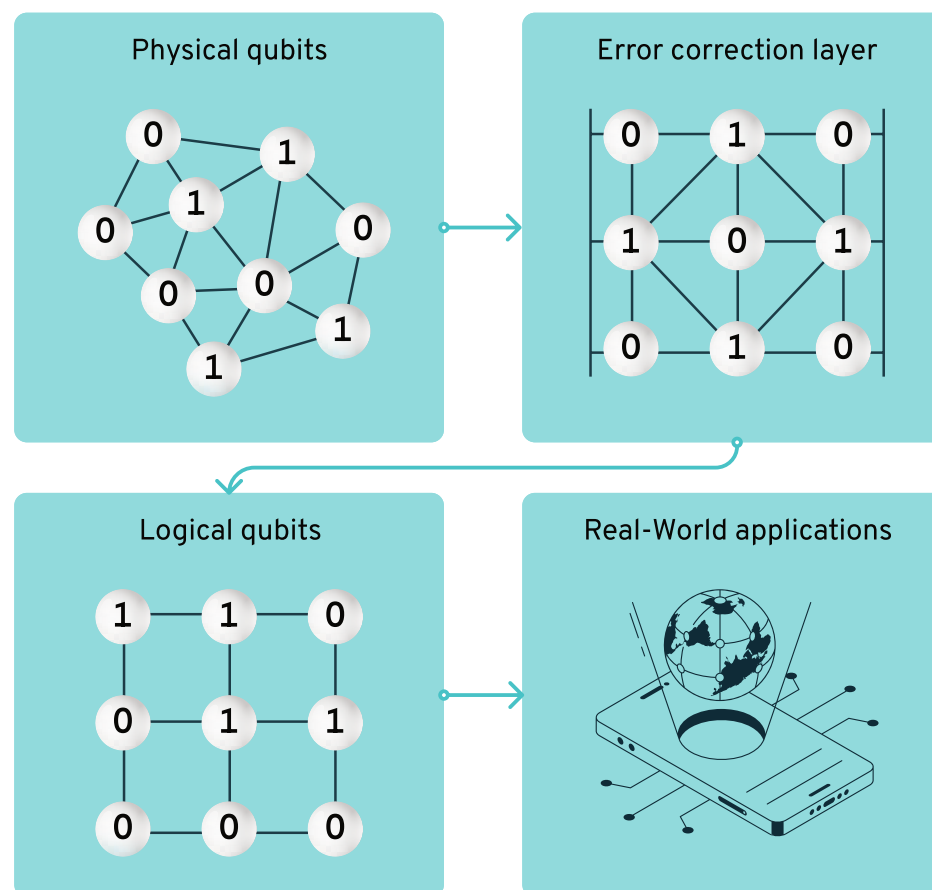
"The first thing I would look for is a deep focus within the company on building fault-tolerant quantum technologies. That doesn't mean they can't have short-term priorities, but I'd want to see a clear commitment at the core—an explicit vision that in five years, or whatever the timeline, they see themselves delivering fault-tolerant systems."

Stephen Bartlett

Professor at the University of Sydney

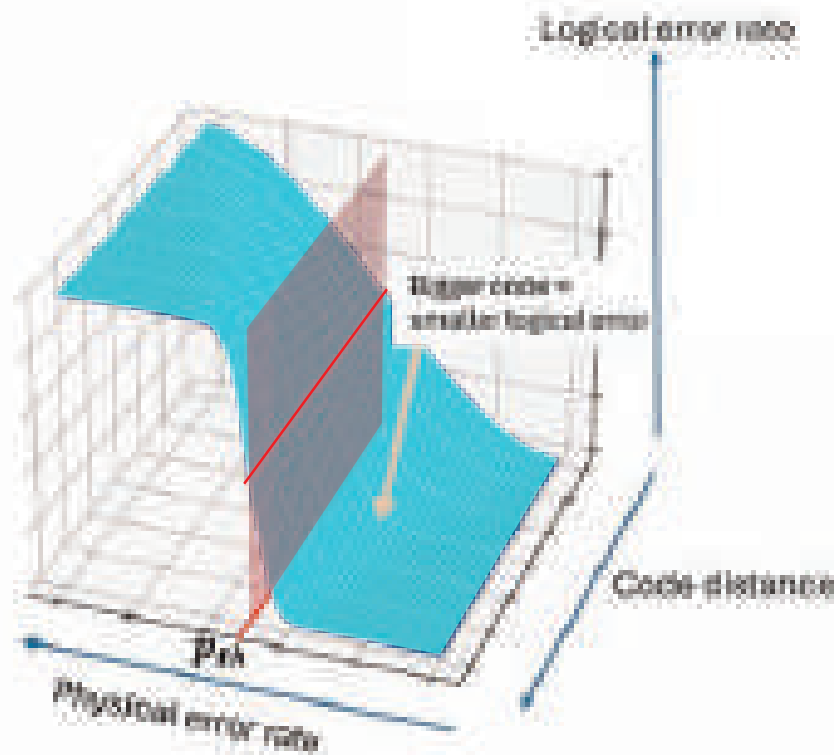
FTQC refers to a quantum computer's ability to execute arbitrarily long and complex computations reliably, despite the fact that every physical step in a quantum circuit, be it flipping a qubit, reading out a result, or correcting for an earlier mistake, has a chance of introducing errors due to the fragile nature of quantum states. This is achieved by encoding logical qubits into multiple physical qubits using quantum error correction (QEC) and ensuring that all operations are performed in a way that detects and corrects errors. As Peter Shor, who helped lay the theoretical foundations of the field, writes:

*"The goal of fault-tolerant quantum computation is to compute accurately in the presence of noise. We assume that each elementary quantum gate, measurement, or storage step may be subject to a small probability of error, and we construct methods which make it unlikely that these errors accumulate in a way that corrupts the final result."*⁵



From fragile qubits to reliable machines through quantum error correction.

This captures the essential idea of FTQC: that even imperfect hardware can be used to compute reliably, provided errors are actively managed through carefully designed circuits and correction protocols.



Quantum error correction becomes scalable when the physical error rates of a system fall below a specific threshold.

Crucially, quantum error correction becomes scalable when the physical error rates of a system fall below a specific threshold. Below this threshold, scaling the code reduces the logical error rate, allowing computations to grow longer and more complex without being overwhelmed by noise. This scaling effect means that adding more physical qubits to protect a logical qubit can push logical error rates down exponentially, making reliable computation more feasible even with imperfect hardware.⁶

This highlights the essence of FTQC: scaling requires not just more qubits, but logical error correction at unprecedented depth. In practice, achieving the large-scale quantum advantage required for algorithms such as integer factorization or quantum chemistry simulation demands logical error rates in the range such that hundreds to thousands of physical qubits are necessitated per logical qubit.⁷

For example, factoring RSA-2048, a benchmark often used to illustrate the scale of the problem, was once thought to require tens of millions of physical qubits. Recent analyses have reduced this estimate to under one million, yet even this lower figure highlights the extraordinary technical demands of building systems capable of sustaining such large-scale computations.⁸

Introduction



"The real challenge is scale—scaling up. And I think that's broadly recognized, but it doesn't diminish how significant a problem it is. The costs and resource overheads of making logical qubits out of physical qubits, and implementing fault-tolerant quantum logic from many, many physical gates, are just incredibly high. It will require large numbers of qubits and very deep circuits."

Stephen Bartlett

Professor at the University of Sydney



"If they can reliably hit 10^{-3} as an average error rate, with a very very tight distribution of error rates across all qubits, on fundamental gates in an integrated device with a dozen or so qubits all operating at the same time, then I would say focus on integration and control and figure out how to get a price-per-qubit below \$1, when averaged over all elements of the hardware system."

Simon Devitt

Professor at the University of Technology, Sydney



"We do indeed need to incorporate error-corrected logical qubits, error correction in order to be able to run longer calculations to reach utility scale. That pushes the timeline out because of the large overhead and numbers of qubits and operational control. And so the challenges are controlling more qubits, reducing error rates, showing that error correction can be implemented with a degree of control and fidelity that gives us these very deep quantum calculations with billions of gate operations."

Mark Saffman

Chief Scientist for Quantum Information at Infleqtion



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Qubit Modalities

As of 2025, several qubit modalities are considered mature, widely researched, and commercially pursued. These include superconducting circuits, trapped ions, neutral atoms, semiconductor spin qubits, and photonic qubits. All have demonstrated high-fidelity gates and there are active error correction efforts. Each provides distinct advantages and faces unique scaling challenges.

Matter-based qubits (superconducting, trapped ion, neutral atom, spin) benefit from long-lived states and deterministic gates. Photonic qubits, by contrast, operate at room temperature, naturally support long-distance transmission, and are well-suited to measurement-based quantum computing (MBQC).

“For strictly qubit modalities it is still tough to displace the surface code, primarily due to the difficulty in the underlying hardware supporting more exotic codes like constant rate qLDPC. The error-correction community has historically had waves of surface code replacements in the past 20 years, with post-selected QEC and gauge-fixed color codes as two notable examples. In principle they have superior performance to the surface code, but ultimately they are very difficult to engineer on a device. The current qLDPC wave is the best that has come along in a while, but it remains to be seen if hardware chipsets can be built supporting the long range connectivity at error rates suitably below threshold for large-scale operations.”

Simon Devitt

Professor at the University of Technology, Sydney.

While each modality has distinct advantages and equally specific limitations, there is growing interest in the potential for hybrid quantum computing architectures that integrate complementary technologies to overcome individual bottlenecks and enhance overall system performance.⁹

Two-Axis Classification of Qubit Modalities

To better understand the landscape of qubit technologies and their fault-tolerance strategies, it is useful to classify platforms along two fundamental axes: the physical nature of the qubit carrier and the computational model it supports. This dual perspective helps illuminate both the constraints and opportunities shaping each modality's path to scalability.

Physical Nature

Quantum hardware platforms can be grouped into two broad categories: matter-based and photon-based systems. Matter-based systems include trapped ions, neutral atoms, superconducting circuits, spin qubits, and donor qubits, where qubit states are encoded in physical degrees of freedom like spin, charge, or energy levels of trapped particles.

Photon-based systems, as their name suggests, use photons as qubit carriers and rely on optical components and detectors for gate operations and measurements.

Computational Model

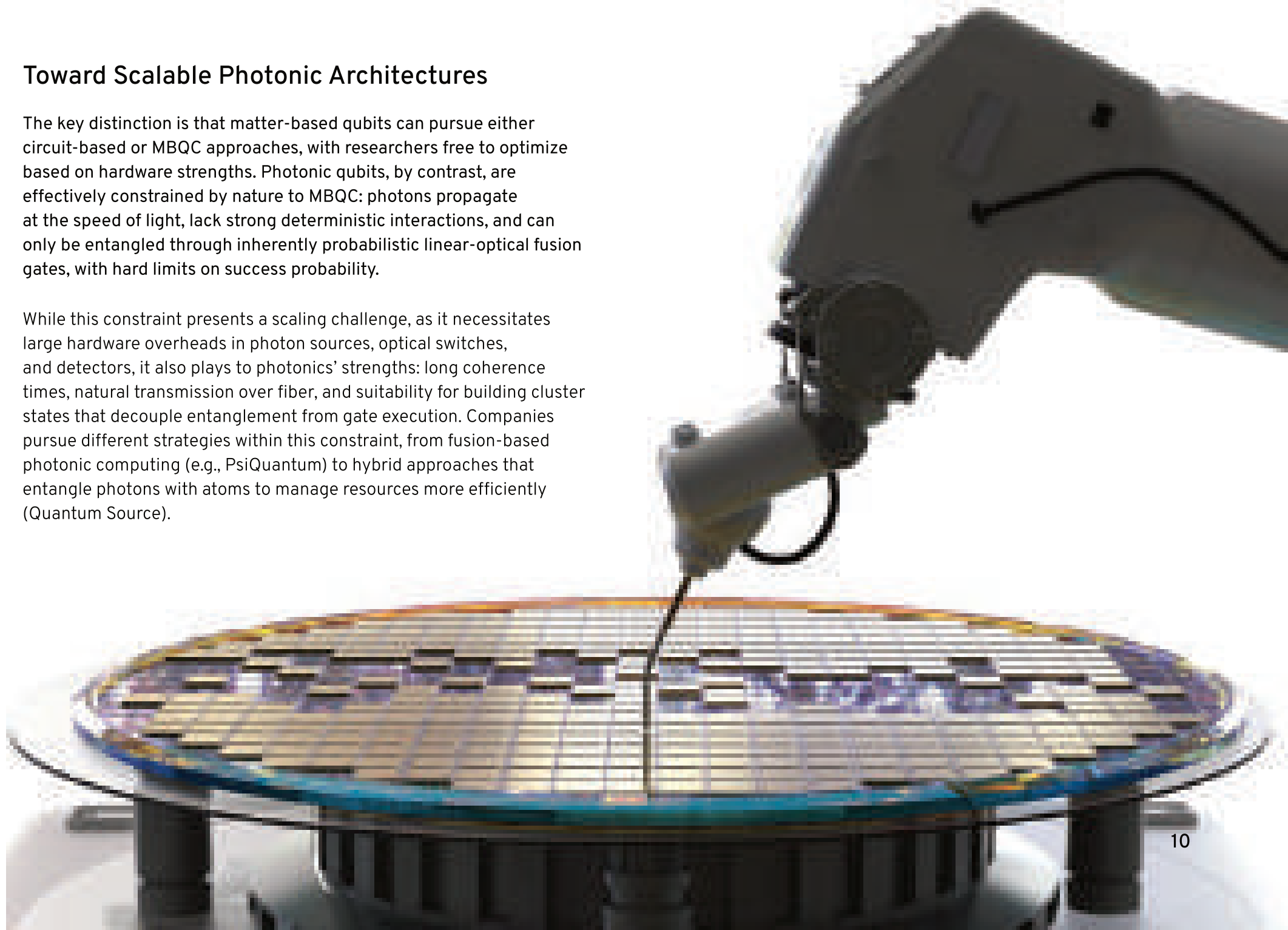
There are two primary computational models. In the circuit-based model, quantum algorithms are implemented as a series of single- and two-qubit gates applied directly to qubits, analogous to classical digital circuits. Both matter-based and photonic qubits can, in principle, support this model. In practice, however, circuit-based photonics is impractical at scale: photons are “flying” qubits that propagate at the speed of light, so holding them through many sequential gates requires long delay lines or memories, and every pass through an optical element adds loss. Coupled with the lack of strong deterministic two-photon interactions, these losses compound and make large-scale circuit implementations in photonics untenable with current technology.

In the measurement-based quantum computing (MBQC) model, computation is carried out through adaptive single-qubit measurements performed on a pre-prepared entangled cluster state, with the choice of each measurement basis guided by the outcomes of earlier ones¹⁰. This approach separates entanglement generation from logical execution and mitigates the storage/interaction problem for photons: probabilistic entangling operations can be performed “offline,” and photons need only be routed and measured with low-latency feed-forward rather than stored for long gate sequences.

Toward Scalable Photonic Architectures

The key distinction is that matter-based qubits can pursue either circuit-based or MBQC approaches, with researchers free to optimize based on hardware strengths. Photonic qubits, by contrast, are effectively constrained by nature to MBQC: photons propagate at the speed of light, lack strong deterministic interactions, and can only be entangled through inherently probabilistic linear-optical fusion gates, with hard limits on success probability.

While this constraint presents a scaling challenge, as it necessitates large hardware overheads in photon sources, optical switches, and detectors, it also plays to photonics' strengths: long coherence times, natural transmission over fiber, and suitability for building cluster states that decouple entanglement from gate execution. Companies pursue different strategies within this constraint, from fusion-based photonic computing (e.g., PsiQuantum) to hybrid approaches that entangle photons with atoms to manage resources more efficiently (Quantum Source).





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Matter-Based Qubit Platforms

Matter-based qubits encode quantum information in long-lived states of physical systems such as atoms, superconducting circuits, or spins. These platforms support deterministic two-qubit gates through direct interactions, providing a foundation for scalable logic. Leading modalities include superconducting circuits, trapped ions, neutral atoms, and semiconductor spin qubits.

Key performance benchmarks include gate speed (how quickly operations can be executed, from nanoseconds in superconductors to microseconds in ions and atoms), coherence times (T_1 : energy relaxation and T_2 : dephasing, indicating how long quantum information can be preserved), fidelity (the accuracy of single- and two-qubit gates), and connectivity (the pattern in which qubits can interact, from nearest-neighbor in 2D superconducting qubit arrays to all-to-all in ion traps).

“
Qubit connectivity is an industry-wide challenge. If I need the qubit in the upper left to perform a logic gate with the qubit on the lower right, how do I do that? I think everyone suffers from that challenge.”

Mark Saffman

Chief Scientist for Quantum Information at Infleqtion

Quantum error correction strategies for matter-based systems generally focus on surface codes and related topological codes implemented on two-dimensional lattices of physical qubits. Together, these parameters determine the feasibility of implementing large-scale error correction and ultimately achieving fault tolerance.

Approaches to Computation in Matter-Based Platforms

Matter-based platforms can pursue either the circuit-based or measurement-based computational model. The circuit-based approach is dominant: superconducting qubits and semiconductor spins implement algorithms through sequences of single- and two-qubit gates, benefiting from fast gate speeds and growing integration density.

Trapped ions and neutral atoms, while also capable of circuit-based logic, can implement MBQC by generating large, deterministic cluster states and then applying adaptive measurements. This approach leverages their high entanglement fidelity and long storage times, though circuit-based methods may remain more resource-efficient for many algorithms.

Hybrid architectures are also emerging, where matter-based MBQC nodes are connected by photonic interconnects, enabling distributed quantum processing while maintaining deterministic local interactions^{4,9}.

Representative Efforts in Matter-Based Qubits

- **Superconducting qubits**

IBM, Google, Amazon and others scaling planar superconducting arrays toward logical-qubit demonstrations using surface and qLDPC codes.

- **Neutral atoms**

QuEra, Pasqal, and Infleqion developing large 2D/3D arrays with programmable connectivity and long coherence times.

- **Trapped ions**

Quantinuum and IonQ focusing on modular systems with photonic interconnects for scaling beyond hundreds of qubits.

- **Semiconductor spins**

Intel, Diraq, and academic teams integrating spin qubits into CMOS-compatible fabrication flows for ultra-dense scaling.

Matter-Based Qubit Platforms: Superconducting Qubits

Category	Typical Values / Characteristics
Physical basis / encoding	Macroscopic states in circuits containing Josephson junctions (transmons, flux qubits, fluxoniums)
Operating conditions	Dilution refrigerators, ~10–20 mK
Gate times	Single-qubit: ~20 ns; Two-qubit: ~40 ns
Fidelities	Single-qubit: 99.998% ¹¹ ; Two-qubit: 99.94% ¹²
Coherence times	>100 μ s typical; >1 ms research ³⁹
Connectivity	Nearest-neighbor on 2D chip arrays with long-range connections
QEC progress	Surface-code logical error suppression ¹ ; qLDPC proposed and simulated with 12 logical qubits in 288 physical ^{14,41}
Scalability advantages	Mature fabrication ecosystem; ultra-fast gates
Key challenges	Fabrication yield, cryogenic wiring/control, long-range connectivity

Overview

Superconducting qubit circuits are fabricated on chips and cooled to ~10–20 mK. Within these specially engineered circuits, specific designs (such as transmons, flux qubits, and fluxoniums) give rise to distinct quantum states that serve as qubits, encoded in the quantized energy levels of the superconducting circuit, whose nonlinearity is provided by Josephson junctions.

These systems leverage mature semiconductor fabrication, microwave control hardware, and very fast gate times, making them one of the most developed qubit modalities. Compared to other platforms, superconducting circuits have among the shortest gate times (two-qubit gates ~40 ns) and a robust technology ecosystem¹. Their main noise sources are dielectric loss, magnetic flux fluctuations, and thermal excitations, which limit coherence and gate fidelity¹⁵.

Current State

- Coherence times (T_1 , T_2) now routinely exceed 100 μ s, with research devices surpassing 1ms⁴⁰.
- Gate fidelities have reached extremely high levels: single-qubit gates at ~99.998%¹¹, and two-qubit CNOT gates at ~99.94%¹².
- Industry milestones include IBM's scaling of superconducting arrays (127 to 1,121 qubits) and Google's "Willow" processor (105 qubits) demonstrating exponential suppression of logical error rates with increasing code distance¹.

Fault-Tolerance & QEC

Superconducting platforms are natural candidates for the 2D surface code, which has a relatively high error threshold (around 1%) and only requires nearest-neighbor gates⁷. Recent demonstrations include Google's small surface-code "patches" and IBM's qLDPC simulations, which promise similar error protection with far fewer qubits¹⁴. Experimental work on bosonic codes, such as cat qubits and GKP states, is also progressing¹⁶.

Scalability Challenges

Scaling superconducting qubits to millions of units faces several tightly coupled engineering hurdles:

- **Fabrication yield and uniformity:**

Current devices already suffer from variability across wafers; for example, small shifts in Josephson junction parameters lead to large frequency and coherence variations. At the million-qubit scale, maintaining uniform device performance and reproducibility across fabrication runs becomes a bottleneck.

- **Integration & modularity:**

Achieving large-scale FTQC will likely require multi-chip integration or modular architectures. Interconnect yield, coupling efficiency, and synchronization between modules are all active research problems.

- **Cryogenic control infrastructure:**

Each qubit today requires multiple microwave control lines running into a dilution refrigerator at around 10–20 mK.

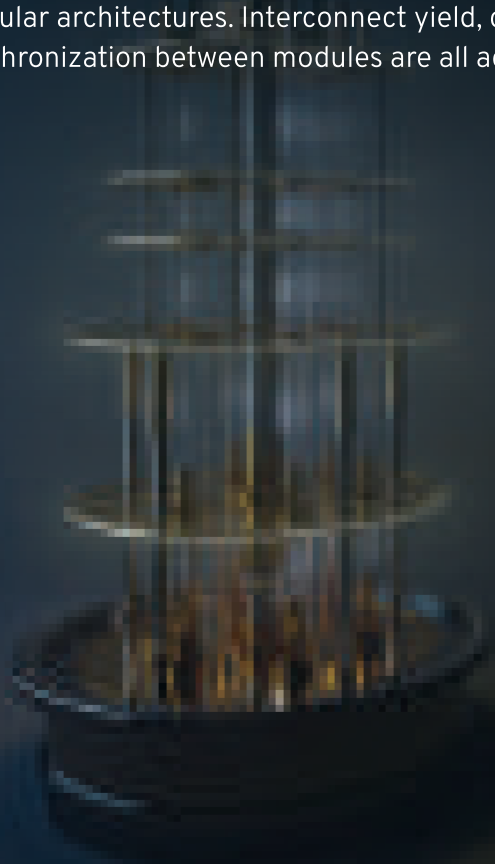
At the million-qubit level, this becomes physically and thermally untenable. Research into cryo-CMOS or superconducting electronics and multiplexed wiring aims to reduce cabling density and power load, but remains at early stages.

- **Crosstalk and microwave crowding:**

As qubits are packed more densely, unwanted coupling between neighboring control lines and microwave frequencies grows, degrading gate fidelities. Advanced packaging and error-mitigation strategies will be required.

- **Physical qubit overhead for QEC:**

Surface codes require thousands of physical qubits per logical qubit. Even if qLDPC codes reduce overhead by an order of magnitude, systems still demand millions of physical qubits for useful algorithms. This raises challenges in fabrication scale, cooling, and system integration.



Matter-Based Qubit Platforms: Trapped-Ion Qubits

Category	Typical Values / Characteristics
Physical basis / encoding	Hyperfine or Zeeman states of ions ($^{171}\text{Yb}^+$, $^{40}\text{Ca}^+$, $^{133}\text{Ba}^+$) in electromagnetic traps
Operating conditions	Ultra-high vacuum; laser or microwave manipulation; often cryogenic temperatures
Gate times	Single-qubit: 10–100 μs ; Two-qubit: 10–100 μs
Fidelities	Single-qubit > 99.99% ¹⁷ ; Two-qubit: $\geq 99.9\%$ ¹⁸
Coherence times	>10 minutes with dynamical decoupling ¹⁹
Connectivity	All-to-all within ion chains via motional modes
QEC progress	First true QEC break-even; logical gates outperforming physical gates ²
Scalability advantages	Identical qubits; flexible connectivity
Key challenges	Scaling ion chains beyond hundreds; photonic interconnect efficiency; complex laser infrastructure

Overview

Trapped-ion qubits encode information in long-lived internal states of atomic ions (commonly $^{171}\text{Yb}^+$, $^{40}\text{Ca}^+$, $^{133}\text{Ba}^+$), confined in electromagnetic traps. Quantum information is typically stored in hyperfine or Zeeman levels, offering exceptional coherence times (seconds to minutes)¹⁹. Qubits are manipulated using laser or microwave pulses that couple internal states to collective motional modes.

A key advantage is uniformity: all ions of a given species are identical, eliminating fabrication mismatches. Furthermore, ions provide reconfigurable connectivity, meaning any pair of ions in a chain can be directly entangled without swap operations. For example, an 11-qubit processor demonstrated full connectivity in a $^{171}\text{Yb}^+$ chain²⁰, and high-fidelity entanglement across arbitrary pairs in a 17-ion system was achieved²¹.

Current State

- Single-qubit fidelities up to 99.9999% have been reported¹⁷; two-qubit fidelities of 99.9%+ are routine in both academic and industry systems¹⁸.
- Hyperfine qubits demonstrate >10 minutes of coherence under dynamical decoupling.
- IonQ and Quantinuum systems currently operate 36–56 qubits with full all-to-all connectivity.



Fault-Tolerance & QEC

Trapped ions are highly compatible with error correction strategies due to their low physical error rates and flexible connectivity. Surface and color codes can be implemented by segmenting traps or using ion shuttling, while qLDPC codes particularly benefit from the platform's all-to-all qubit connectivity, which reduces the overhead required for encoding.

Importantly, trapped ions have already reached several major milestones in fault tolerance. In 2024, Quantinuum's H2 system demonstrated true break-even quantum error correction where logical qubits outperformed physical qubits by suppressing errors hundreds of times more effectively²². Building on this, in 2025 researchers demonstrated a fault-tolerant non-Clifford gate (controlled-Hadamard) whose logical error rate was lower than that of its physical implementation². These advances indicate that trapped ions may require fewer overhead qubits than other modalities, with estimates suggesting only tens to hundreds of physical qubits per logical qubit, compared to the thousands typically needed in superconducting systems²³.

Scalability Challenges

Despite excellent qubit quality, scaling trapped ions to FTQC faces engineering barriers:

- **Mode crowding in long ion chains:**

As the number of ions in a trap grows, vibrational modes become densely packed. This slows down gates, increases crosstalk, and makes calibration harder. Chains beyond 100 ions require sophisticated error mitigation or alternative architectures.

- **Photonic interconnects:**

Linking ion modules via remote entanglement (ion-photon interfaces) is essential for scaling to millions of qubits. However, photon collection efficiency, mode-matching, and entanglement rates remain too low for practical modular architectures.

- **Need for multi-zone traps:**

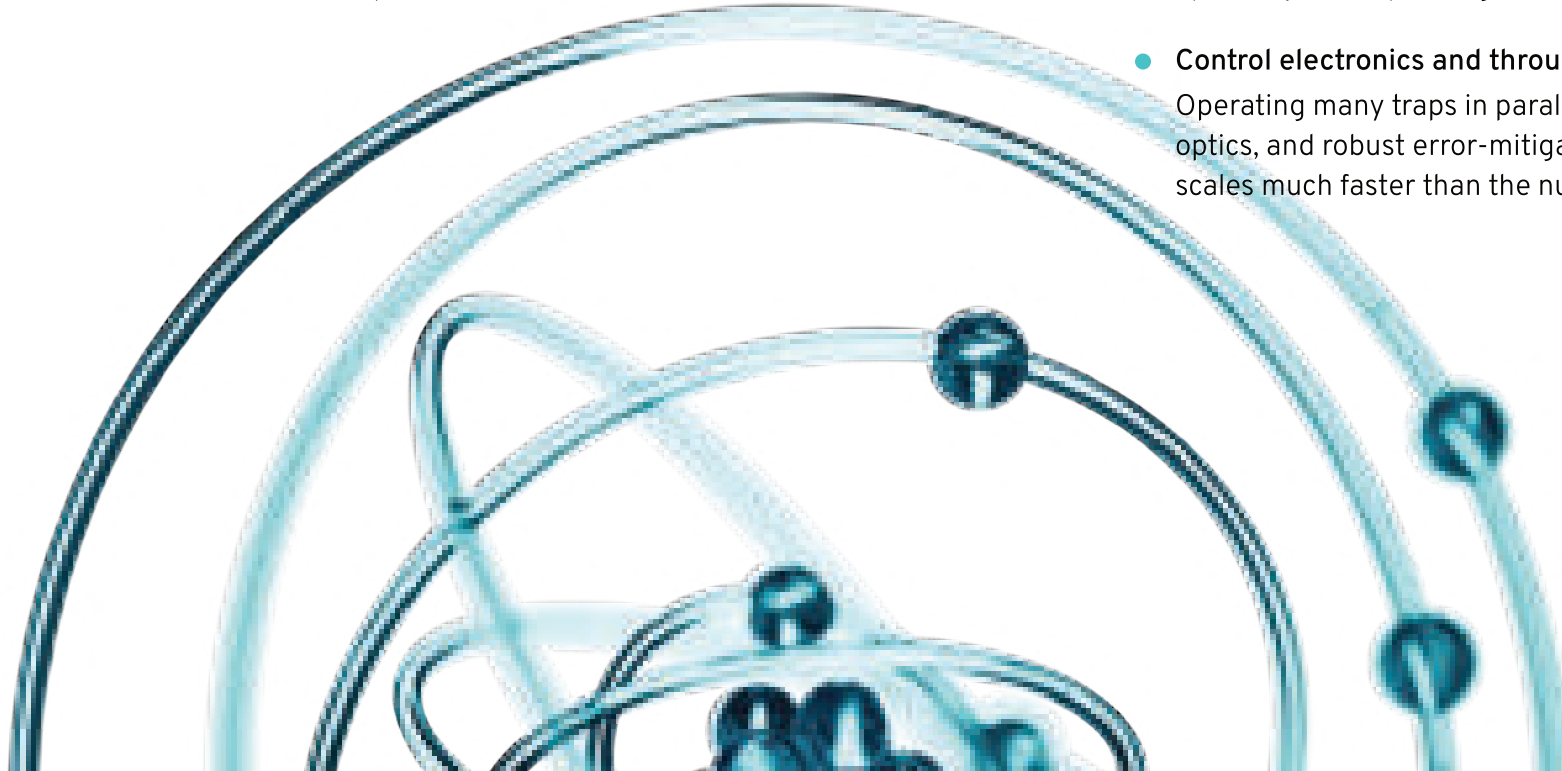
To scale beyond hundreds of ions, researchers must develop large, segmented traps where ions can be shuttled between zones. Building such traps with high yield and stability is technically challenging.

- **Laser infrastructure complexity:**

Each ion requires stable lasers for cooling, state preparation, and entangling gates. A system with thousands of ions would demand hundreds of precisely stabilized laser beams and advanced optical systems, pushing the limits of current engineering.

- **Control electronics and throughput:**

Operating many traps in parallel requires ultra-fast electronics, scalable optics, and robust error-mitigation pipelines. The control complexity scales much faster than the number of qubits.



Matter-Based Qubit Platforms: Neutral-Atom Qubits

Category	Typical Values / Characteristics
Physical basis / encoding	Atomic ground states (Rb, Cs, Sr, Yb); entangling gates via Rydberg excitations
Operating conditions	Optical tweezer/lattice arrays; room temperature mostly
Gate times	Single-qubit: $\sim \mu\text{s}$; Two-qubit (Rydberg): few μs
Fidelities	Single-qubit: $>99.9\%$ (up to 99.97%); Two-qubit: $99\text{--}99.7\%$ ^{24,25}
Coherence times	Up to $\sim 12.6\text{ s}$ ³
Connectivity	Reconfigurable all-to-all via atom rearrangement
QEC progress	Multi-round surface-code error correction in 288-atom arrays; erasure-biased thresholds up to $\sim 10\%$ ²⁶
Scalability advantages	Defect-free large arrays; flexible geometry; parallelism
Key challenges	Raising two-qubit fidelities to $>99.9\%$; laser stability; advanced control for $>10^4$ qubits; speed of operations (shuttling and readout)

Overview

Neutral-atom qubits encode quantum information in the internal states of individual atoms, typically alkali or alkaline-earth elements such as rubidium, cesium, strontium, or ytterbium. Atoms are held in place using tightly focused optical tweezers or arranged in optical lattices, providing highly uniform qubits that are effectively “defect-free” by design. Entangling operations are achieved by exciting atoms to strongly interacting Rydberg states, enabling two-qubit gates on the microsecond timescale.

A distinguishing strength of neutral-atom systems is their reconfigurable geometry: atoms can be repositioned to entangle arbitrary pairs, yielding flexible, large-scale all-to-all connectivity²⁷. These features make neutral-atom platforms particularly attractive for scaling quantum processors into the tens of thousands of qubits.

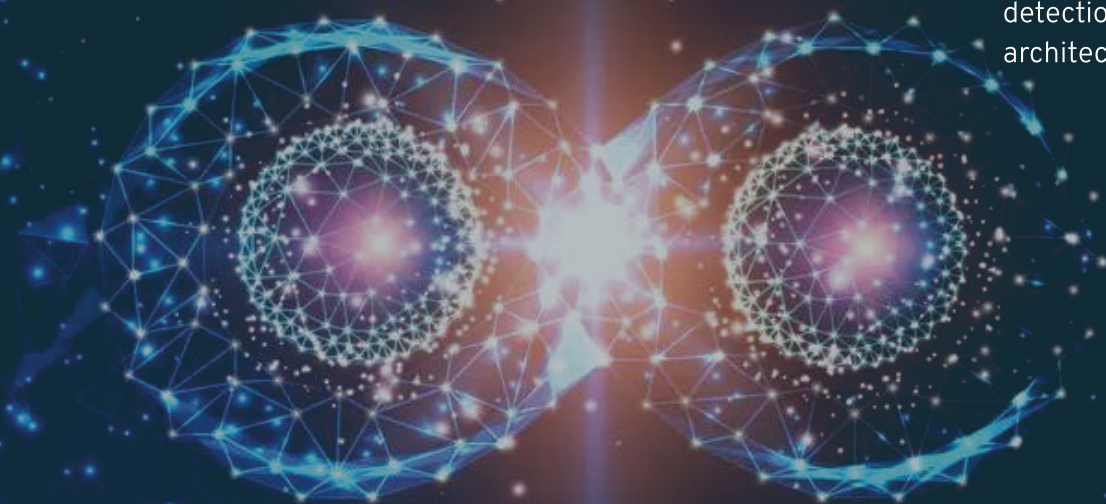
Current State

- **Scale:** Arrays of 6,100+ atoms trapped in optical tweezers³; defect-free 2D/3D arrays of up to 2,024 atoms assembled in 60 ms¹²
- **Coherence:** Hyperfine qubits with coherence times up to 12.6s; atom lifetimes ~23 minutes in optical traps³.
- **Gate fidelity:** Single-qubit >99.9% (up to 99.97% with Raman control²⁴; two-qubit Rydberg gates 99–99.7%²⁵.
- **Parallelism:** Entangling gates demonstrated across tens of atoms; ongoing work targets 99.9% two-qubit fidelity.

Fault-Tolerance & QEC

Neutral-atom platforms are naturally suited for error correction due to their scalability, reconfigurable connectivity, and compatibility with surface or other topological codes. Unlike many platforms limited by stochastic gate noise, neutral atoms exhibit erasure-dominated errors (e.g., atom loss), which can actually be advantageous. Recent theoretical work shows that such erasure-biased noise raises error thresholds well above 1%, with optimized fusion schemes achieving thresholds near 10%²⁶.

Experimentally, progress has been rapid. In 2025, a collaboration between QuEra, Harvard, MIT, Caltech, and UMD demonstrated sustained rounds of error correction on arrays of up to 288 atoms, implementing transversal gates, lattice surgery, and logical teleportation. Recently, they have demonstrated below threshold performance on surface code⁴². Because of their parallelism and reconfigurability, neutral-atom platforms may also support error detection circuits that require fewer time steps than conventional architectures.



Scalability Challenges

Despite scalable assembly, neutral-atom QEC faces major hurdles. Gate fidelities and stability must improve: roadmap goals include >99.9% two-qubit gates and cryogenic cooling to reduce noise. Furthermore, Rydberg gates are sensitive to laser intensity/phase noise, and operating arrays of $>10^4$ atoms will demand advanced control hardware.

Nonetheless, neutral atoms' compact qubit footprint ($\sim 1\text{ }\mu\text{m}$ traps) and compatibility with photonic interconnects (e.g. optical fiber links) make them strong candidates for very large systems. If engineers meet these challenges, neutral atoms could become a premier FTQC platform. In the near term, companies like QuEra (cold-atom quantum computing) and academic labs (Harvard, Innsbruck, others) are rapidly advancing system size and gate fidelity on Rydberg arrays.

Despite their promise, neutral-atom qubits face several critical hurdles on the path to large-scale FTQC:

- **Gate fidelity and stability:**

Pushing two-qubit Rydberg fidelities from 99.5–99.7% toward 99.9%+ is essential. This requires better laser coherence, power stabilization, and error-mitigation protocols.

- **Shuttling and readout speeds:**

Current scale-up approaches for neutral atom-based systems rely on mechanical “shuttling” of atoms, which takes tens to hundreds of microseconds. Readout speeds are currently in the range of 100 μs to 1 ms; this also constitutes a major bottleneck and necessitates massively parallel operation.

- **Integration with photonic interconnects:**

Neutral atoms are naturally compatible with optical fiber links, but efficient atom–photon interfaces for modular architectures are still in early stages.

- **Laser and optical infrastructure:**

Operating arrays of thousands of atoms demands highly stable, multi-wavelength laser systems with precise intensity and phase control. Scaling these systems while keeping complexity and cost manageable remains a challenge.

- **Cryogenic operation for noise suppression:**

While neutral-atom qubits operate at room temperature, cryogenic cooling may be required for suppressing blackbody radiation effects on Rydberg states at very large scales.

- **Classical control overhead:**

Parallel manipulation of 10,000+ atoms will require advanced optical and electronic control systems with high-bandwidth feedback.

Matter-Based Qubit Platforms: Semiconductor Spin Qubits

Category	Typical Values / Characteristics
Physical basis / encoding	Electron/nuclear spins in silicon quantum dots or dopant atoms
Operating conditions	Dilution refrigerators, ~10–100 mK (some designs higher cryo temps)
Gate times	Single- and two-qubit: ~ μ s
Fidelities	Single-qubit: up to 99.98%; Two-qubit: $\geq 99.5\%$ ^{29,30}
Coherence times	Up to ms in isotopically enriched silicon
Connectivity	Nearest-neighbor in arrays; extendable via exchange couplings
QEC progress	Three-qubit phase-flip code demonstrated ³¹ ; universal gate set above threshold ³⁰
Scalability advantages	CMOS-compatible fabrication; nanoscale footprint (millions per chip possible)
Key challenges	Device variability; cryo-CMOS control; wiring density at scale

Overview

Semiconductor spin qubits encode information in the spin of electrons or nuclei confined in quantum dots or dopant atoms. Their microscopic size (~10–100 nm) makes them uniquely suited for ultra-high-density integration, with the prospect of millions of qubits per chip. Fabrication using established CMOS processes offers a direct path to leveraging the semiconductor industry's scaling expertise.

These qubits operate at cryogenic temperatures, typically 10–100 mK, though some designs work at higher cryogenic ranges, potentially easing cooling requirements. Historically, spin qubits lagged behind in gate speed and control complexity, but rapid advances are closing this gap.

Current State

- **Gate fidelities:** First demonstration of fault-tolerance-compatible fidelities in 2022: 99.8% single-qubit, 99.5% two-qubit²⁹.
- **Industrial fabrication:** Intel fabricated spin qubits on 300-mm wafers with >99% fidelities³².
- **Universal gate sets:** Diraq achieved a fully universal, calibration-free gate set with fidelities above surface-code thresholds³⁰.
- **Coherence times:** Millisecond-scale in isotopically enriched silicon; further extended with dynamical decoupling.

Fault-Tolerance & QEC

Spin qubits' combination of long coherence and high fidelity enables error correction. Early demonstrations include a three-qubit phase-flip code protecting against a single-bit phase error in silicon³¹.

Their main QEC challenge is scale: current devices involve just a handful of qubits, while fault tolerance requires hundreds per logical qubit. Nonetheless, CMOS-compatible fabrication and cryo-CMOS control electronics offer a roadmap for integrating large-scale QEC circuits. In principle, spin qubits could match or even outperform other solid-state modalities once these engineering bottlenecks are addressed.

Scalability Challenges

Scaling spin qubits to millions of devices presents both opportunities and barriers:

- **Fabrication uniformity:**
While CMOS compatibility is an advantage, quantum-dot qubits are highly sensitive to nm-scale variations in device geometry, which can produce large variability across arrays.
- **Wiring density:**
Each qubit requires multiple control electrodes; wiring thousands to millions of devices at cryogenic temperatures risks routing congestion. Multiplexed readout and cryo-CMOS electronics are under active development.
- **Gate speed:**
Microsecond gate times, while not prohibitive, demand optimized control sequences to minimize QEC overhead compared to faster superconducting platforms.
- **Limited demonstrations of QEC:**
Unlike superconducting and ion platforms, spin qubits are at an earlier stage in multi-qubit QEC experiments. The first large-scale demonstrations of logical error suppression remain to be shown.



Photonic Qubits and Hybrid Architectures

Photonic qubits encode quantum information in the properties of individual photons rather than in material states. These properties include polarization, time-bin, frequency, or spatial mode. Photonic qubits operate natively at room temperature, without the need for cryogenic refrigeration or ultra-high vacuum trapping, are intrinsically resilient to many forms of environmental noise, and can transmit quantum states over long distances with very low decoherence.

The primary challenge is that photons lack strong deterministic interactions, making high-fidelity two-qubit gates difficult to implement directly. As a result, most linear-optical photonic architectures, such as those pursued by PsiQuantum, adopt measurement-based quantum computing (MBQC), where entanglement is generated probabilistically and consumed via adaptive measurements. Representative approaches include fusion-based photonic processors, continuous-variable encodings, and hybrid atom-photon systems.

Key performance metrics include photon-source efficiency, optical loss, detector efficiency, and feed-forward latency. For fusion-based approaches, photon indistinguishability is also critical. Together, these determine the feasibility of constructing large-scale cluster states and executing fault-tolerant protocols.



Measurement-Based Quantum Computing in Photonics

For photonic systems, measurement-based quantum computing (MBQC) is the natural architectural model. Because photons propagate at the speed of light, they cannot be easily halted for sequential circuit-based gate operations. MBQC instead embeds logic into the topology of pre-prepared cluster states, allowing computation to proceed via adaptive measurements.

The lack of strong native photon-photon interactions also makes direct two-qubit gates impractical. MBQC overcomes this by concentrating probabilistic entangling operations into an offline resource-generation stage. At the same time, MBQC leverages existing photonic strengths: efficient photon sources, low-loss routing, and high-performance detectors can be optimized to build scalable cluster-state “factories” with low-latency feed-forward control.

Fusion-based MBQC is a leading model, in which smaller entangled states are probabilistically fused into large three-dimensional cluster states, protected by loss-tolerant topological codes that can tolerate up to ~25% photon loss¹⁰. Continuous-variable MBQC represents another promising direction, employing large-scale squeezed-state cluster states.

Representative Efforts in Photonic Qubits

- **PsiQuantum** – Pursuing large-scale fusion-based MBQC on silicon photonics, targeting one million physical qubits. Architecture relies on high-efficiency photon sources, ultra-low-loss switches, and integration of superconducting nanowire single-photon detectors (SNSPDs).
- **Xanadu** – Developing continuous-variable MBQC via time-domain multiplexing of squeezed states, demonstrated with >200 modes.
- **Quantum Source’s Atom-Photon Entanglement Approach** – Uses atomic quantum memories entangled with photons to create hybrid matter-light cluster states, enabling deterministic storage and synchronization of photonic qubits.
- **University of Bristol / USTC spins** – Demonstrations of large-mode photonic cluster states with active feed-forward.

Photonic Qubits and Hybrid Architectures: Photonic Qubits

Category	Typical Values / Characteristics
Physical basis / encoding	Single photons encoded in polarization, time-bin, path, or frequency
Operating conditions	Room temperature; fiber or free-space propagation
Gate times	Determined by cluster-state measurement speed and feed-forward
Fidelities	Sources: >99% purity; indistinguishability ~97–98% ^{33,34}
Coherence times	Essentially decoherence-free in transmission
Connectivity	Long-distance via optical fibers; cluster-state entanglement via fusion gates
QEC progress	Loss-tolerant topological codes in MBQC; fusion-based cluster-state demos; logical gate teleportation
Scalability advantages	Room-temperature operation; natural networking and modularity
Key challenges	Photon loss; probabilistic entanglement; ultra-low-loss optics; ultrafast feed-forward

Overview

Photonic qubits use single photons—or, in some approaches, squeezed light modes—as carriers of quantum information. States can be encoded in polarization, time-bin, path, or frequency. Photons are highly robust and experience negligible decoherence over long distances and operate at room temperature, making them natural candidates for quantum communication and distributed architectures.

However, photons lack strong deterministic interactions. As a result, photonic quantum computing generally relies on measurement-based quantum computing (MBQC). Large entangled cluster states are generated upfront, and computation proceeds via adaptive single-photon measurements [Raussendorf, 2001]. Common platforms span bulk optics experiments, integrated photonic chips (e.g. silicon nitride, lithium niobate), and hybrid systems that couple photons to matter-based memories.

Current State

- **Single-photon sources:** Deterministic quantum-dot emitters achieve purity >99% and indistinguishability up to ~97.5%^{33,34}.
- **Detectors:** State-of-the-art SNSPDs approach 100% efficiency³⁵ with timing jitter down to 11 ps³⁶.
- **Integrated photonics:** Programmable interferometers with >1,000 elements and <0.1 dB loss per component have been demonstrated³⁷.
- **Large-scale systems:** PsiQuantum has recently begun constructing large-scale photonic quantum computing facilities in Brisbane and Chicago, targeting operational readiness by the end of 2027. Its approach leverages mature semiconductor fabrication and telecom technologies to pursue a million-qubit, fault-tolerant quantum computer, positioning PsiQuantum as the only major company focused on achieving fault-tolerance through a photonic architecture. Xanadu has recently demonstrated the first on-chip error-resistant (GKP) photonic qubits and introduced *Aurora*, a modular networked photonic architecture designed for scalable error correction.

Fault-Tolerance & QEC

Photonic FTQC efforts focus on loss-tolerant MBQC architectures. Topological cluster states can tolerate up to ~25% photon loss, making them well suited to photonics¹⁰. Early demonstrations include teleportation-based fusion of encoded photonic states and logical gate teleportation on integrated chips.

Fusion-based quantum computation is a leading strategy: small entangled “resource states” are probabilistically generated and stitched together into large-scale cluster states. This approach reduces overhead compared to direct large-cluster generation and aligns with integrated photonic hardware. Importantly, MBQC consumes photons as the computation progresses—unlike matter-based circuit models—placing additional emphasis on ultra-efficient source and detector technology.

Scalability Challenges

Scaling photonic quantum computing faces several core obstacles:

- **Probabilistic entanglement and resource overhead:**
In linear optics, two-photon entangling gates succeed only probabilistically, requiring extensive resources to approach near-deterministic performance. Practical cluster states demand enormous numbers of synchronized photons, switches, and detectors. This overhead motivates integration with quantum memories or hybrid matter-based systems to buffer and synchronize resources.
- **Photon loss:**
Even minimal per-component losses accumulate across large optical circuits (e.g., 0.1 dB loss per element becomes prohibitive across 1,000+ elements).
- **Feed-forward speed:**
MBQC requires ultrafast, real-time adjustments to measurement bases. Achieving sub-nanosecond feed-forward with large photonic arrays demands advanced low-latency electronics.

Hybrid and Atom-Photon Approaches

Overview

Hybrid quantum architectures combine the strengths of matter-based and photonic systems. A central idea is to use matter qubits (such as atoms or solid-state emitters) as deterministic mediators of photon-photon entanglement. In this “atom-as-interface” approach, single atoms are coupled to high-finesse optical resonators, enabling them to emit photons on demand and implement entangling gates between them. This bypasses the probabilistic limitations of linear-optical schemes by providing deterministic photon sources and high-fidelity matter-photon gates.

“There’s likely to be a modular architecture where we stitch modules together, either with motion or with light, probably with light. But if each module can be as large as possible, that will be a benefit.”

Mark Saffman

Chief Scientist for Quantum Information at Infleqion

Recent advances highlight the potential of this strategy. For instance, Quantum Source introduced a cavity-mediated scheme in which atoms act as deterministic mediators for generating and stitching large photonic graph states. The results, supported by detailed simulations, show how atom-photon interfaces could overcome the probabilistic limits of linear-optical fusion by enabling repeatable, high-fidelity operations³⁸.

“With our technology, we are able to scale our qubit counts to at least 100,000 physical qubits within a single vacuum chamber. To scale further, we plan to use atom-photon interconnects in a modular fashion. Neutral atom technology is a natural fit to support these interconnects as atom-photon interactions are already the basis of every operation in the system. In anticipation of this transition, we are actively developing advanced techniques for high-bandwidth communications between modules.”

Dr. Ben Bloom

Founder & CEO at Atom Computing

By shifting from probabilistic to deterministic entanglement, such hybrid schemes could reduce the resource overheads associated with photonic FTQC, bringing cluster-state architectures closer to practical scalability.

Beyond photon-photon logic, hybrid systems are also being explored as interconnects: photons serve as carriers linking spatially separated matter-based processors (e.g., trapped-ion or superconducting modules). Hybrid approaches combine the high-fidelity operations of local matter qubits with the networking advantages of photonics, making them strong candidates for modular, distributed FTQC architectures.

Quantum Source Case Study

Quantum Source occupies a distinct position in the photonic quantum computing landscape. Unlike other photonic players—such as PsiQuantum, which relies on probabilistic fusion-based entanglement—Quantum Source integrates atomic qubits directly into the entanglement process, enabling deterministic generation of large photonic cluster states.

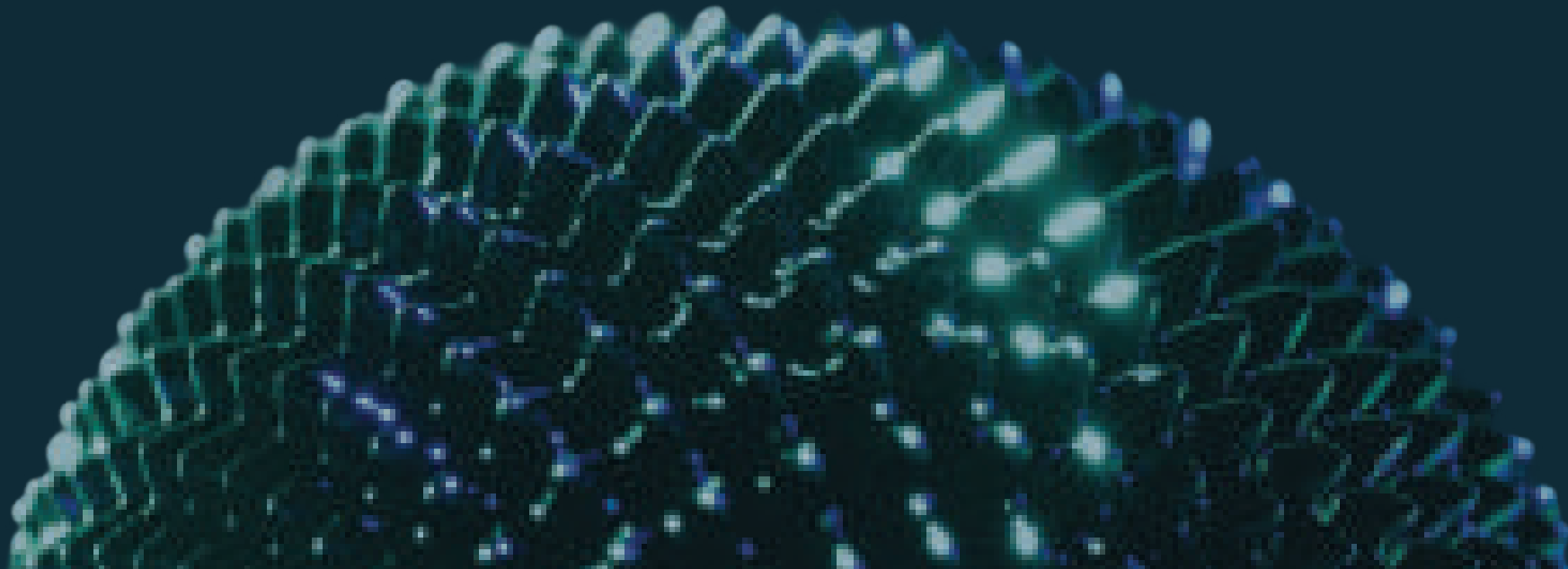
Position in the Two-Axis Framework

- **Physical nature:** Light-based (photonic) qubits.
- **Computation model:** Measurement-based quantum computing (MBQC).
- **Core differentiation:** Deterministic atom–photon entanglement, reducing resource demands compared to probabilistic-only approaches.

Technical Approach

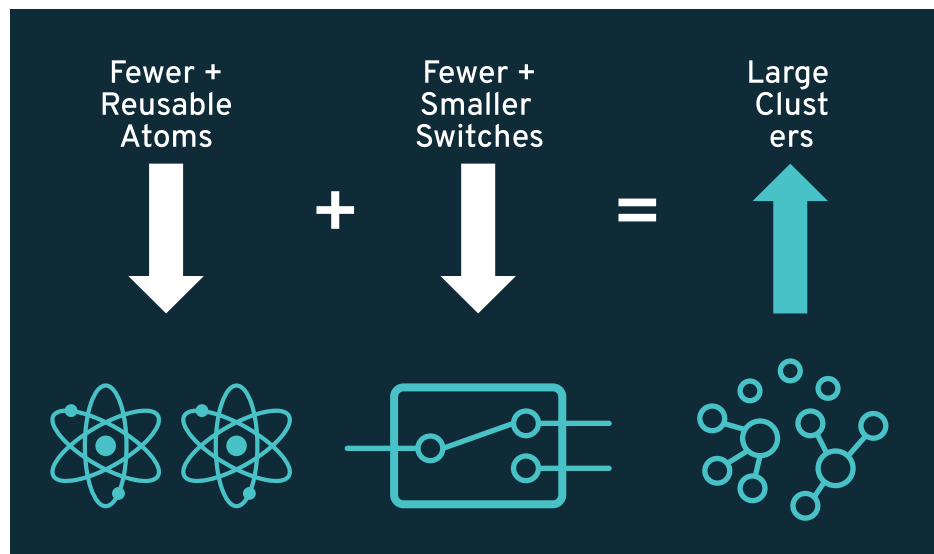
In standard photonic MBQC, resource state generators (RSGs) produce small entangled photon clusters probabilistically. Because these entangling operations succeed only rarely, achieving fault tolerance requires millions of synchronized photon sources, intricate optical switching networks, and extensive cryogenic infrastructure. This enormous resource overhead is a fundamental bottleneck to scalability. **Quantum Source circumvents this by using trapped atoms as intermediary entanglement nodes:**

- A single photon is entangled with an atom using cavity quantum electrodynamics (cQED) techniques.
- The atomic state is then mapped to a second photon, entangling the two photons indirectly through the shared atomic state.
- The same atom can be recycled for repeated entanglement operations, conserving costly atomic resources while producing large entangled photonic states efficiently.



Advantages Over Conventional Photonic MBQC

- **Scalability:** Deterministic entanglement eliminates the exponential overhead of probabilistic fusion gates.
- **Resource Efficiency:** Significantly fewer single-photon sources and optical switches are required, reducing system complexity and footprint.
- **Integration Potential:** Compatible with room-temperature photonic systems, reducing cryogenic dependencies.
- **Flexibility:** Atomic resources are conserved and reused, while photons—cheap and abundant—form the bulk of the computational qubits.



Resource efficiency underpins Quantum Source's scalability advantage.

This architecture directly addresses the primary photonic bottleneck of two-photon entanglement by sidestepping photon-photon gates in favor of a reliable atom-photon interaction. The result is a platform that maintains photonics' long-distance and room-temperature benefits while dramatically improving fault-tolerance feasibility.

“
By harnessing deterministic photon-atom interactions on a chip, we can generate entangled photonic states with unprecedented efficiency – at room temperature, in a compact and scalable architecture – paving the way for practical and cost-effective quantum computers.”

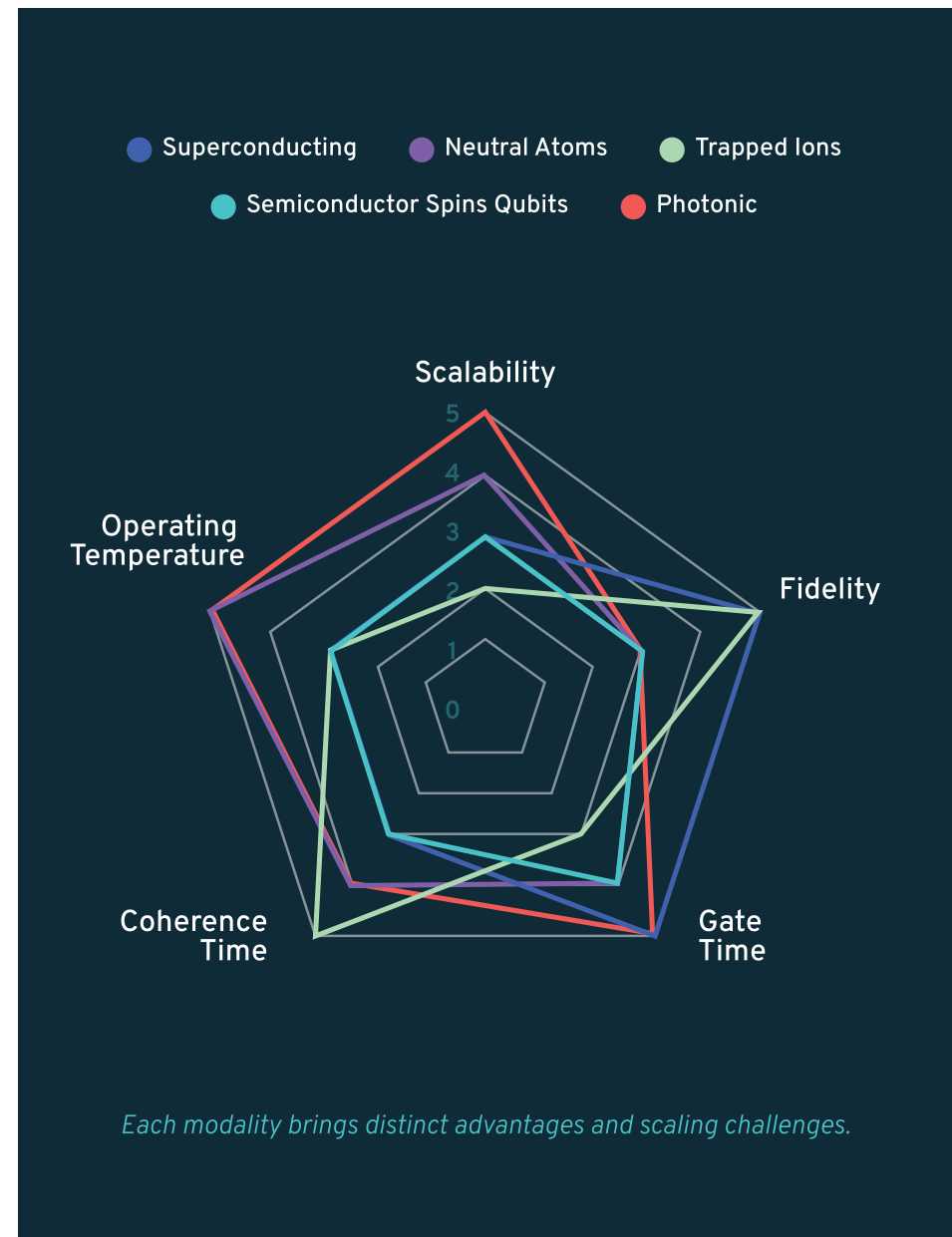
Oded Melamed
CEO at Quantum Source

Conclusion

Across modalities, the path to fault-tolerant quantum computing is defined by balancing physical performance, error-correction overhead, and engineering scalability.

Comparative Assessment

- **Superconducting Qubits:** Mature fabrication, ultra-fast gates, and high fidelity with sub-threshold logical qubits; however, high cryogenic and wiring overheads pose significant scalability challenges.
- **Trapped Ions:** Exceptional coherence and fidelity with all-to-all connectivity; however, scaling is challenged by mode crowding, slower gates, and complex laser control.
- **Neutral Atoms:** Flexible connectivity, large-array scaling, global control, and multi-second coherence; improving two-qubit fidelity and addressing relatively slow shuttling and readout speeds remain critical targets for FTQC.
- **Semiconductor Spins :** CMOS compatibility and microscopic footprint, but sensitive to variability and reliant on cryogenic integration, limiting scalability.
- **Photonic Qubits:** Room-temperature operation, natural networking capability, and long-distance coherence enable scalable architectures; however, photon loss, probabilistic entanglement, and feed-forward requirements remain key challenges.



The Hybrid Opportunity

Quantum Source's deterministic atom–photon approach addresses photonics' scaling bottleneck, replacing probabilistic entanglement with high-fidelity deterministic links. This strategy not only strengthens photonics as a standalone modality but also enables modular architectures that connect matter-based processors over photonic networks.

Outlook

Current progress suggests complementary strengths across modalities: superconductors in integration, ions in fidelity, neutral atoms in scale, spins in density, and photons in networking. While hybrid approaches that merge these capabilities remain largely untested, they could provide a viable route for creating heterogeneous systems where matter-based processors could eventually be linked by photonic interconnects.

Within the next decade, we expect logical qubits across multiple modalities to consistently outperform physical qubits, and million-qubit systems to become a realistic engineering target. For investors and policymakers, the message is clear: supporting a diversity of hardware—particularly hybrid innovations—will accelerate the breakthrough to practical, error-corrected quantum computers.

The task ahead is challenging: reduce errors, scale architectures, and co-design hardware and software for robust QEC. Yet the trajectory is unmistakable. FTQC is no longer a distant aspiration but an approaching reality, with transformative implications for science, industry, and society.

“ Fault tolerance needs to be at the core of every serious company’s roadmap—not just in terms of thinking about today’s architecture, but in considering how those architectures should evolve as systems grow. What’s required is true co-design of the architecture and the underlying technology as companies move along the road to fault tolerance. ”

Stephen Bartlett

Professor at the University of Sydney

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